

Georg von Westphalen

Stay
fluffy!

MedKayfab: Pharmacology Wrestling

Uncle FrizzleBob's
Krazy Kayfabizzaro Komik
Kards for Narrative
Medical Learning Games

BLÖDSINN!



Diagnosis?!



Roxy Receptor

Receptor Theory

Kd (Dissociation Constant)

POWER: Affinity & Efficacy: Lower K_d means a tighter grip.

LORE: Response is proportional to fraction of receptors occupied, but spare receptors mean you don't need 100% to win.



Georgie GPCR

G-Protein Coupled Receptors

34% of all approved targets



POWER: The Transducer: Ligand binds $\rightarrow$ Ga-GTP dissociates $\rightarrow$ downstream chaos.

LORE: Ga_s activates adenylyl cyclase. Ga_i inhibits it. Ga_q triggers IP3/DAG.

KayfaBINGO!



KayfaBINGO!

Kinny Kinase

Receptor Tyrosine Kinases

1 Transmembrane Domain



POWER: The Dimer-Strike: Ligand binding forces dimerization and explosive downstream phosphorylation.



LORE: Drives insulin and growth factors. Primary target for oncology kinase inhibitors.

Nicky Nuclear

Nuclear Receptors

Hours to Days (Onset)



POWER: Permanent Ink: Changes gene expression directly. Slow to act, slow to fade.

LORE: Explains why corticosteroids take hours to work and why abrupt withdrawal causes HPA axis collapse.

Iggy Ion Channel

Ligand-Gated Ion Channels

Milliseconds
(Speed)

POWER: Lightning Strike: Ligand binding physically opens the pore. Fastest receptor in the game.

LORE: Targets include nAChR (muscle contraction), GABA-A (benzos), and NMDA (ketamine).



Volta the Voltage-Gated

Action Potential Targets

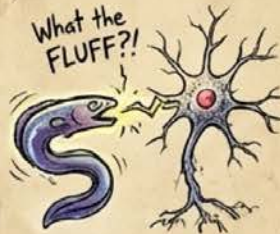
pH < 7.4
(Kills Local Anesthetics)

POWER: Use-Dependent Block: Local anesthetics only bind when the channel is firing and in the inactivated state.

LORE: Local anesthetics are weak bases. In acidic, infected tissue, they ionize outside the cell and fail to penetrate.



What the FLUFF?!



Beta the Beta-Arrestin

100% Downregulation Desensitization & Tolerance

POWER: Signal Termination: Uncouples receptors that have been partying too long.

LORE: Explains tachyphylaxis (asthma inhaler overuse) and opioid tolerance via receptor internalization.



Second the Messenger

cAMP, IP3/DAG, cGMP

PDE
(The Degradator)

POWER: The Relay: Carries the signal from the receptor to the cellular machinery.

LORE: Sildenafil inhibits PDE5, stopping the degradation of



KayfaBOGGLE!

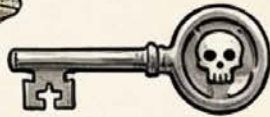
How can a right-handed molar & a left-handed molar - the same molar - do different things?

Allo the Allosteric

Allosteric Modulation



Emax Preserved

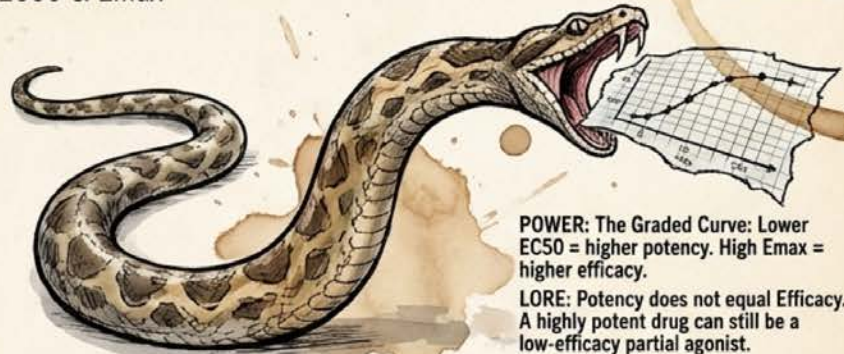


POWER: The Side-Door Hack: Binds away from the active site to change affinity or efficacy of the main agonist.

LORE: Benzodiazepines are positive allosteric modulators at GABA-A, they only work in the presence of GABA, ensuring a safety ceiling.

Dose the Dose-Response

EC50 & Emax



EC50 (Potency) vs. Emax (Efficacy)

POWER: The Graded Curve: Lower EC50 = higher potency. High Emax = higher efficacy.

LORE: Potency does not equal Efficacy. A highly potent drug can still be a low-efficacy partial agonist.

Abby Absorption

Bioavailability - F



$F = \text{AUC(oral)} / \text{AUC(IV)}$

POWER: The Gauntlet: The fraction of administered dose reaching systemic circulation unchanged. IV is 100%.

LORE: Highly lipophilic, un-ionized drugs absorb best. Sublingual and transdermal bypass the liver entirely.

First Pass Frank

First-Pass Metabolism



>70% Extraction Ratio

POWER: Portal Vein Toll: Hepatic metabolism destroys high-extraction drugs before they reach the bloodstream.

LORE: Explains why GTN must be sublingual and lidocaine IV. Prodrugs exploit this to become active.

Vera

Volume of Distribution

$V_d > 100L$
(Tissue Bound)

POWER: Tissue Sink: High V_d means the drug hides in fat/muscle, making plasma levels a lie.

LORE: Amiodarone is so extensively tissue-bound that dialysis cannot remove it in an overdose.



Protein Priya

Plasma Protein Binding

Only Free Drug Acts

POWER: The Reserve: Bound drug is inactive, cannot cross membranes, and cannot be excreted.

LORE: In liver disease, the free fraction of highly bound drugs skyrockets, causing toxicity at normal levels.



Hal the Half-Life

$t_{1/2}$ & Steady State

$5 \times t_{1/2} = \text{Steady State}$

POWER: The Decay: Time to eliminate half the drug. Depends on both Clearance and Volume of Distribution.

LORE: It takes 5 half-lives to reach steady state, regardless of dosing frequency.



Zero the Zero-Order

Non-Linear Kinetics

Constant Rate Elimination

POWER: Enzyme Saturation: Eliminates a fixed AMOUNT per unit time, regardless of concentration.

LORE: Phenytoin and alcohol follow zero-order kinetics. A small dose increase can cause a catastrophic spike.

HUMBUG!



Clara Clearance

Hepatic & Renal CL

eGFR < 30
(Metformin CI)



POWER: The Washout: The volume of blood cleared of drug per unit time.
LORE: Drugs with high renal clearance require strict dose reduction in renal failure to prevent toxic accumulation.

Cyp the CYP450 Chief

Metabolism

~50%
(CYP3A4 share)



POWER: The Monooxygenase: The superfamily of enzymes that decides the fate of your drugs.
LORE: CYP2C9 metabolises active S-warfarin. CYP2D6 metabolizes codeine. CYP3A4 metabolizes half of everything.

Indie the Inducer

Enzyme Induction

Days to Weeks
(Onset)



POWER: Gene Upregulation: Forces the liver to build more enzymes, destroying substrate drugs faster.
LORE: Rifampicin = OCP = unintended pregnancy. Takes days to weeks to start and stop.

Inhi the Inhibitor

Enzyme Inhibition

Immediate
Effect



POWER: The Blockade: Competes for the CYP active site, causing substrate drug levels to spike dangerously.
LORE: Statins + CYP3A4 inhibitors = massive rhabdomyolysis risk. Effect is Immediate upon dosing.

Poly the Polymorphism (Pharmacogenomics)

POWER: The Genetic Dice:
DNA variants change enzyme
function from ultra-rapid to
completely broken.

LORE: CYP2D6 poor metabolizers
get zero pain relief from codeine.
Ultra-rapid convert it to
toxic doses of morphine.

~7% Poor
Metabolizers



Fight! Fight! Fight!



Thera the Monitor (Therapeutic Drug Monitoring)

Narrow TI
Only

POWER: The Blood Watch:
Only used for drugs where the
guess would be dangerous and
levels correlate with effect.

LORE: Timing is everything:
Digoxin requires a trough >6h
post-dose; Aminoglycosides
use peak/trough.



Sym the Sympathetic (Fight or Flight)

$\alpha 1$, $\beta 1$, $\beta 2$

POWER: The Adrenaline Rush:
High heart rate, bronchodilation,
vasoconstriction,
and zero digestion.

LORE: Epinephrine
activates all subtypes.
Noradrenaline primarily
hits $\alpha 1$ and $\beta 1$ with
minimal $\beta 2$.



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Para the Parasympathetic (Rest & Digest)

M1, M2, M3

POWER: The Vagal Brake: Slows
heart rate, contracts smooth
muscle, and ramps up secretions
via Acetylcholine.

LORE: M2 slows the heart
(blocked by atropine =
tachycardia). M3 controls
the bladder and gut.



POWER: The Vagal Brake: Slows heart rate, contracts smooth muscle (blocked by atropine = tachycardial). M3 controls the bladder and gut.

Nicky NMJ (Neuromuscular Junction)

SV2A & SNARE targets



POWER: The Muscle Switch: ACh floods the nAChR, causing depolarization and muscle contraction.
LORE: Rocuronium (non-depolarizing) competitively blocks the receptor. Succinylcholine agonizes it, causing flaccid paralysis.

Alpha Annie (Alpha-Adrenoceptor Drugs)

First-Dose Hypotension

POWER: The Vasodilator: Blocking α_1 drops systemic vascular resistance and relaxes prostate smooth muscle.

LORE: Must start α -block BEFORE β -block in pheochromocytoma to prevent hypertensive crisis.



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Stay fluffy!

Beta Barry (Beta-Blockers)

$\beta_1 > \beta_2$ (Cardioselectivity)

POWER: The Cardiac Brake: Reduces HR, contractility, and AV conduction. Reduces post-MI remodeling.

LORE: Non-selective beta-blockers block β_2 , causing deadly bronchoconstriction in asthma. Use cardioselective cautiously.



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LORE: Non-selective beta-blockers block β_2 , causing deadly bronchoconstriction in asthma. Use

Musky the Antagonist (Antimuscarinics)

Tertiary vs Quaternary

POWER: The Drying Agent: Blocks Para's signal. Stops sweating, stops peeing, stops salivating.

LORE: Tertiary amines cross the blood-brain barrier causing confusion in the elderly. Quaternary amines do not.



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LORE: Tertiary amines cross the blood-brain barrier causing confusion in the elderly. Quaternary amines do not.

Cate the Catecholamine (Pressors)

1-3 $\mu\text{g/kg/min}$



POWER: The Emergency Toolkit: Tailored receptor profiles for different types of shock.
LORE: Noradrenaline is first-line for septic shock to restore SVR. Dobutamine provides inotropy for cardiogenic shock.

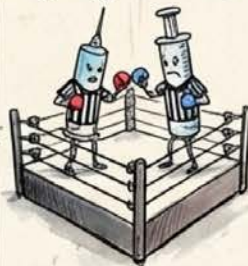
Cholinergic Crisis (Too Much Drug)

Miosis
(Key Sign)



POWER: The Depolarizing Block: Excessive AChE inhibition overloads the NMJ, causing paradoxical weakness.
LORE: Distinguished from Myasthenic crisis at the bedside by muscarinic signs: bradycardia and pinpoint pupils.

Diagnosis?!



Anti the Anticholinesterase (AChE Inhibitors)

SLUDGE
Syndrome



POWER: The Amplifier: Prevents ACh breakdown, amplifying every cholinergic signal in the body.
LORE: Organophosphate poisoning irreversibly blocks AChE. Treat with Atropine and Pralidoxime.

Myasthenic Crisis (Too Little Signal)

Worsening
Weakness



POWER: The Signal Failure: Inadequate ACh effect at the NMJ due to disease progression or infection.
LORE: Treat with IVIg, plasma exchange. Differentiate from Cholinergic crisis by lack of muscarinic signs.



Tole the Tolerance (Adaptation)

24-48
Hours



POWER: The Countermeasure: The body adapts via enzyme induction or receptor downregulation until the drug barely works.
LORE: Continuous GTN infusions cause complete tolerance within 48 hours. A nitrate-free interval is mandatory.

Suzy Synergy (Drug Combinations)

1 + 1 = 3

POWER: The Force Multiplier: Combined effect is greater than the sum of individual parts via complementary mechanisms.

LORE: Co-trimoxazole sequentially blocks two steps of bacterial folate mechanisms.

LORE: Co-trimoxazole sequentially blocks two steps of bacterial folate synthesis, turning two bacteriostatic drugs into a bactericidal synergy.



Anton Antagonist (Competitive vs Non)

Shift Right vs
Drop Down

POWER: The Blockade: Competitive antagonists can be overcome with enough agonist. Non-competitive permanently ruin the lock.

LORE: Naloxone is competitive; a massive fentanyl overdose can outcompete standard naloxone doses.



Thera Therapeutic Index (Safety Margin)

$TI = TD_{50} / ED_{50}$

POWER: The Kill Zone: The ratio between the dose that cures you and the dose that kills you.

LORE: Lithium has a razor-thin TI. Dehydration or NSAIDs easily push normal doses into the seizure toxicity zone.



Indo Induction (General Anesthesia)

POWER: The Immobility Threshold: Lower MAC means a more potent inhalational gas. Lipid solubility drives potency.

LORE: Ketamine preserves respiratory drive and raises heart rate, making it the induction agent of choice in trauma.



MAC

Prot the Prodrug (Metabolic Activation)

POWER: The Disguise: Enters inactive to bypass poor absorption, relying on the liver to forge the active weapon.

LORE: CYP2C19 poor metabolizers cannot activate clopidogrel, leading to catastrophic stent thrombosis.

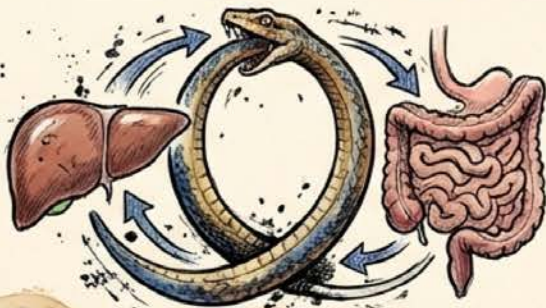


30% Poor Metabolizers

Entero the Enterohepatic (Recycling)

POWER: The Infinite Loop: Conjugated drugs dumped into bile are deconjugated by gut bacteria and reabsorbed.

LORE: Colestyramine interrupts this cycle by binding drugs in the gut, used to urgently wash out teratogens.



Gut flora activation

Preg the Pregnancy Risk (Teratology)

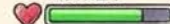
POWER: The Dual Patient: Every drug is a risk-benefit calculation for two. Small lipophilic molecules cross the placenta.

LORE: Valproate causes neural tube defects. ACE inhibitors cause renal agenesis. Warfarin causes bone deformities.

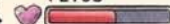


Weeks 6-12 Danger Zone

MOTHER



FETUS



*Alcohol
from and
strongly
contributes
to the
development
of the
fetus.*

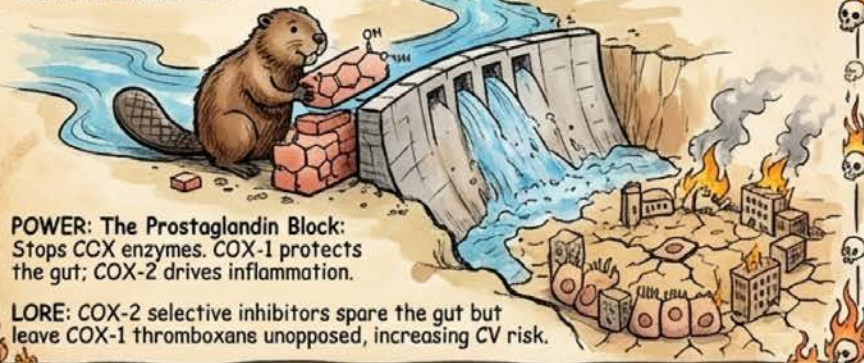
*Ant. New
Horm. HbA1c*

GW

NSAID Neil

COX Inhibition

7-10 Days
(Aspirin)



POWER: The Prostaglandin Block:
Stops COX enzymes. COX-1 protects the gut; COX-2 drives inflammation.

LORE: COX-2 selective inhibitors spare the gut but leave COX-1 thromboxanes unopposed, increasing CV risk.

Proton Petra

PPIs

~18 Hours
(Pump Regen)



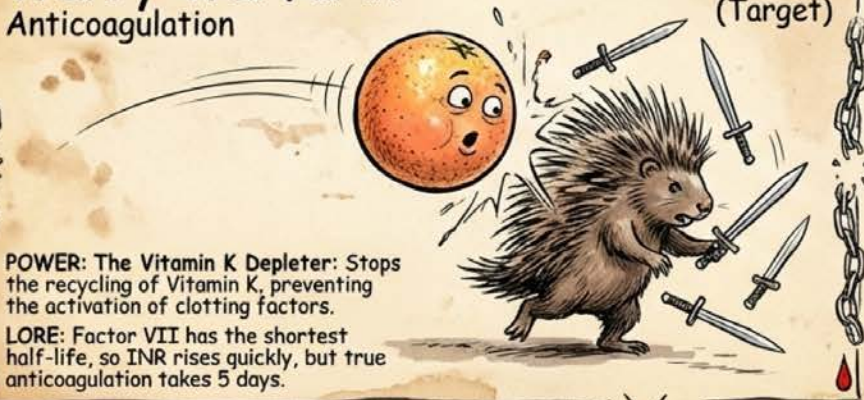
POWER: The Final Common Pathway:
Prodrugs that activate in the acidic canaliculus to irreversibly destroy the proton pump.

LORE: Because the inhibition is irreversible, new pumps must be synthesized. PPIs must be taken before meals.

Warry Warfarin

Anticoagulation

INR 2.0 - 3.0
(Target)



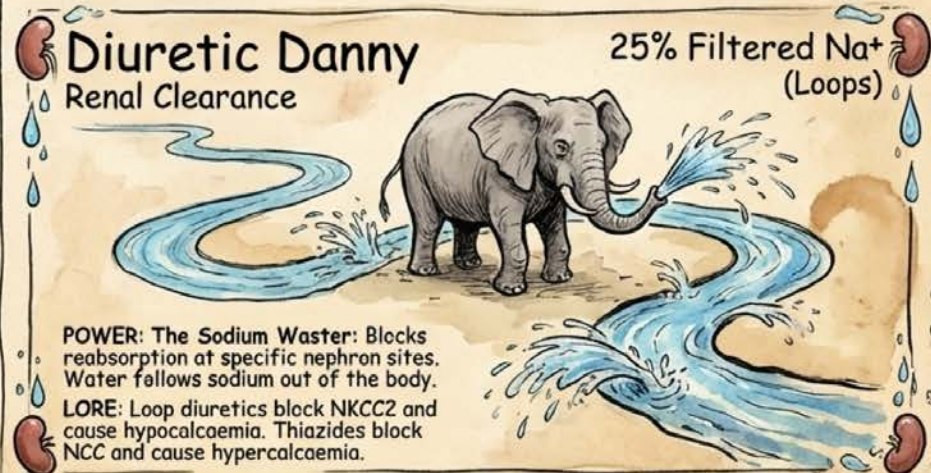
POWER: The Vitamin K Depletter: Stops the recycling of Vitamin K, preventing the activation of clotting factors.

LORE: Factor VII has the shortest half-life, so INR rises quickly, but true anticoagulation takes 5 days.

Diuretic Danny

Renal Clearance

25% Filtered Na⁺
(Loops)



POWER: The Sodium Waster: Blocks reabsorption at specific nephron sites. Water follows sodium out of the body.

LORE: Loop diuretics block NKCC2 and cause hypocalcaemia. Thiazides block NCC and cause hypercalcaemia.

KayfaBIZARRO!



HUMBUG!

Statin Steve

HMG-CoA Reductase

>50% LDL Reduction



POWER: The Receptor Upregulator: Blocking cholesterol synthesis forces the liver to express more LDL receptors.

LORE: Simvastatin is heavily metabolized by CYP3A4; combining with inhibitors drastically increases rhabdomyolysis risk.

Opio the Opioid

Analgesia

10mg IV =
30mg Oral



POWER: The mu-Receptor Agonist: Hyperpolarizes neurons. No ceiling effect for full agonists; endless analgesia and endless respiratory depression.

LORE: Buprenorphine is a partial agonist; if given to a heroin-dependent patient, it violently precipitates withdrawal.

Micro-
stories
— and
unilligible
corners

Antibio the Antibiotic

Mechanisms

5 Distinct
Vulnerabilities



POWER: Selective Toxicity: Exploits the biological differences between prokaryotes and eukaryotes.

LORE: Vancomycin binds the D-Ala-D-Ala terminus to block cell walls. Too large to absorb orally, except for *C. diff*.

Time Kill Terry

PK/PD Targets

T>MIC vs.
C_{max}/MIC



POWER: The Kinetics of Death: Penicillins need constant time above the MIC. Aminoglycosides need one massive peak.

LORE: Aminoglycosides are dosed once daily to maximize the peak while allowing a trough period for renal cell recovery.

Micro-stories + illigible

Micro-stories + illigible

Resi the Resistance (Bacterial Evasion)

ESBL Hydrolyzes Cephs



POWER: Horizontal Gene Transfer: Plasmids share resistance codes (efflux pumps, mutations) across species.

LORE: MRSA alters its PBP2a target so beta-lactams cannot bind. VRE changes D-Ala to D-Lac.

Anti-Viral Vera (Viral Targets)

Selective Activation (TK)



POWER: The Trejan Horse: Nucleoside analogues mimic DNA building blocks, causing chain termination.

LORE: Aciclovir is a harmless prodrug until it is specifically activated by viral Thymidine Kinase.

Azole Aziz (Antifungals)

CYP51 (Demethylase)



POWER: The Sterol Stripper: Fungi use ergosterol; mammals use cholesterol. Azoles inhibit the enzyme that builds it.

LORE: Because azoles are potent CYP inhibitors, they cause massive, fatal interactions with statins and warfarin.

Chemo Clara (Cytotoxics)

S-Phase Specificity



POWER: The Replication Block: Alkylating agents cross-link DNA; antimetabolites fake nucleotides; microtubule agents freeze mitosis.

LORE: Cytotoxics cannot distinguish cancer from healthy fast-growing tissue. Cyclophosphamide causes cystitis.

Stay fluffy!



WAR!

Mutate NOW!



Die Fungi!
No mercy!

KayfaBIZARRO! END OF THE LINE!

Polypharmacy (The Accumulation)



POWER: Cascade Prescribing: Drug A causes side effect, treated by Drug B
LORE: 5+ drugs in the elderly drives falls, cognitive decline, and GI bleed
Deprescribe!

Herb the Herbal (Interactions)



POWER: The Undeclared Guest:
St John's Wort induces CYP3A4;
Grapefruit irreversibly inhibits it.



THE FINAL QUEST:
A 68yo on warfarin, digoxin, and simvastatin gets fluconazole for a fungal infection. Map the interactions. Predict the toxidrome. Show your reasoning.

Remember kid: You remember stories, not lists.
The rules are real, the animals are fake,
and the half-life never lies.
Stay Fluffy

— Doc FrizzleBob

KAYFABIZZARO!